

Q 15 Installation of WordPress and Joomla CMS

A Content Management System (CMS) is software that allows users to build, manage, and publish digital content without requiring in depth knowledge of programming languages such as HTML, CSS, or PHP. This lab documents the installation of two of the most widely used open source CMS platforms : WordPress and Joomla both of which run on a PHP + MySQL/Maria DB stack and are commonly deployed on a local server environment (XAMPP/WAMP/MAMP) for learning purposes before being moved to live hosting.

1.1 Tools and Environment Used

Component	Details
Local Server Stack	XAMPP (Apache, MySQL)
Browser	Brave Browser
WordPress Version	Latest stable release (downloaded from wordpress.org)
Joomla Version	Latest stable release (downloaded from joomla.org)
Database Tool	phpMyAdmin
Text/Code Editor	Visual Studio Code (for reference/customization)

Both WordPress and Joomla are installed following a similar general pattern: set up a local server, create a database, download and extract the CMS files, and run the on screen installation wizard. The detailed steps for each are given below.

1.2 Pre Installation Requirements

- A working local server environment (XAMPP/WAMP) with Apache and MySQL services running.
- PHP version 7.4 or higher (PHP 8.x recommended).
- MySQL 5.7+.
- At least 64 MB of PHP memory limit (128 MB recommended).
- A stable internet connection to download the CMS installation package.

1.3 Installing WordPress

Step by Step Procedure

1. Download WordPress from WordPress.org.
2. Install XAMPP and start the Apache and MySQL modules (if XAMPP is not already installed).
3. Extract the downloaded WordPress ZIP file and copy the "wordpress" folder to the root directory: C:\xampp\htdocs.
4. Open phpMyAdmin: <http://localhost/phpmyadmin>.
5. Create a new database, e.g., `wordpress_db`.
6. Open: <http://localhost/wordpress>.
7. Enter the database details:
Database: `wordpress_db`,
Username: `root`,
Password: blank (default XAMPP),
Host: `localhost`.
8. Enter the site title, admin username, password, and email.
9. Click "Install WordPress".
10. Login through: <http://localhost/wordpress/wp-admin>.

Note: If the installer does not appear automatically, manually create a `wp-config.php` file by copying `wp-config-sample.php` and entering the database credentials, or simply refresh the browser. WordPress will detect the missing configuration and start the wizard.

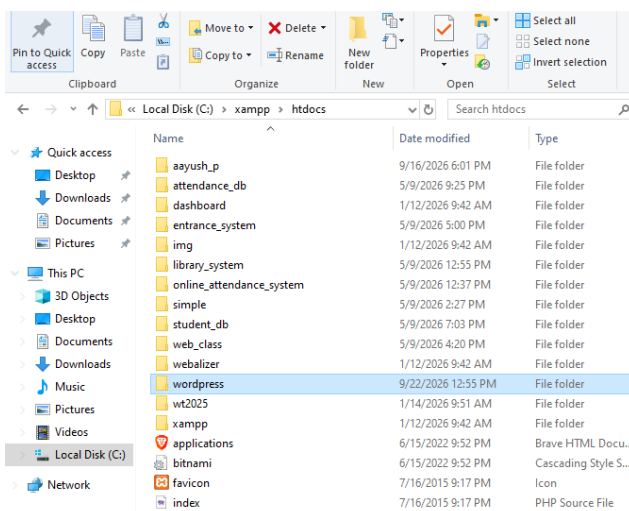


Figure 1: File Explorer showing wordpress

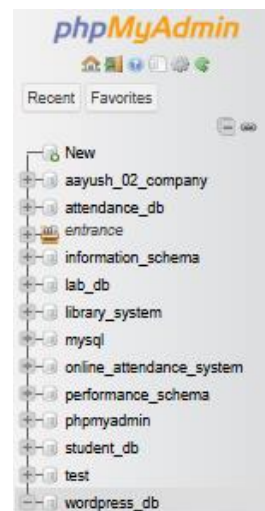


Figure 2: Database of wordpress

WordPress Setup Configuration

localhost/wordpress/wp-admin/setup-config.p...

Below you should enter your database connection details. If you are not sure about these, contact your host.

Database Name:
The name of the database you want to use with WordPress.

Username:
Your database username.

Password: [Show](#)
Your database password.

Database Host:
You should be able to get this info from your web host, if localhost does not work.

Table Prefix:
If you want to run multiple WordPress installations in a single database, change this.

[Submit](#)

Figure 3: Wordpress setup

Dashboard - TechTrail Blog - WordPress

localhost/wordpress/wp-admin/index.php

Howdy, Aayush

Screen Options Help

Welcome to WordPress!

[Learn more about the 7.1.2 version.](#)

[Dismiss](#)

Author rich content with blocks and patterns

Block patterns are pre-configured block layouts. Use them to get inspired or create new pages in a flash.

[Add a new page](#)

Customize your entire site with block themes

Design everything on your site — from the header down to the footer, all using blocks and patterns.

[Open site editor](#)

Switch up your site's look & feel with Styles

Tweak your site, or give it a whole new look! Get creative — how about a new color palette or font?

[Edit styles](#)

PHP Update Recommended

Your site is running on an outdated version of PHP (8.0.30), which does not receive security updates. It should be updated.

Quick Draft

Title

Figure 4: Wordpress dashboard

1.4 Downloading and Installing a WordPress Theme

Step by Step Procedure (from WordPress.org)

1. Visit WordPress.org Themes.
2. Search for the required theme.
3. Open the theme and click "Download".
4. A .zip file will be downloaded.
5. Login to your WordPress Dashboard.
6. Go to "Appearance" → "Themes" → "Add New Theme".
7. Click "Upload Theme".
8. Select the downloaded .zip file.
9. Click "Install Now".
10. Click "Activate" to apply the theme to your website.

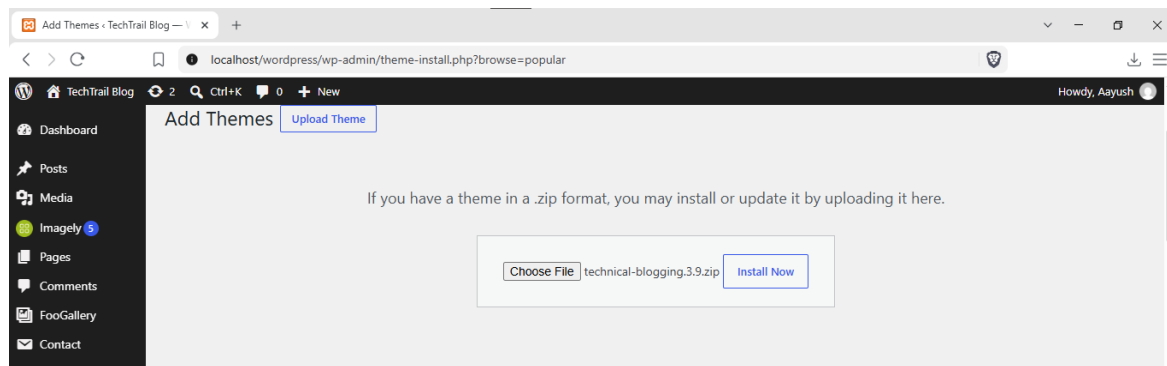


Figure 5: Adding Theme

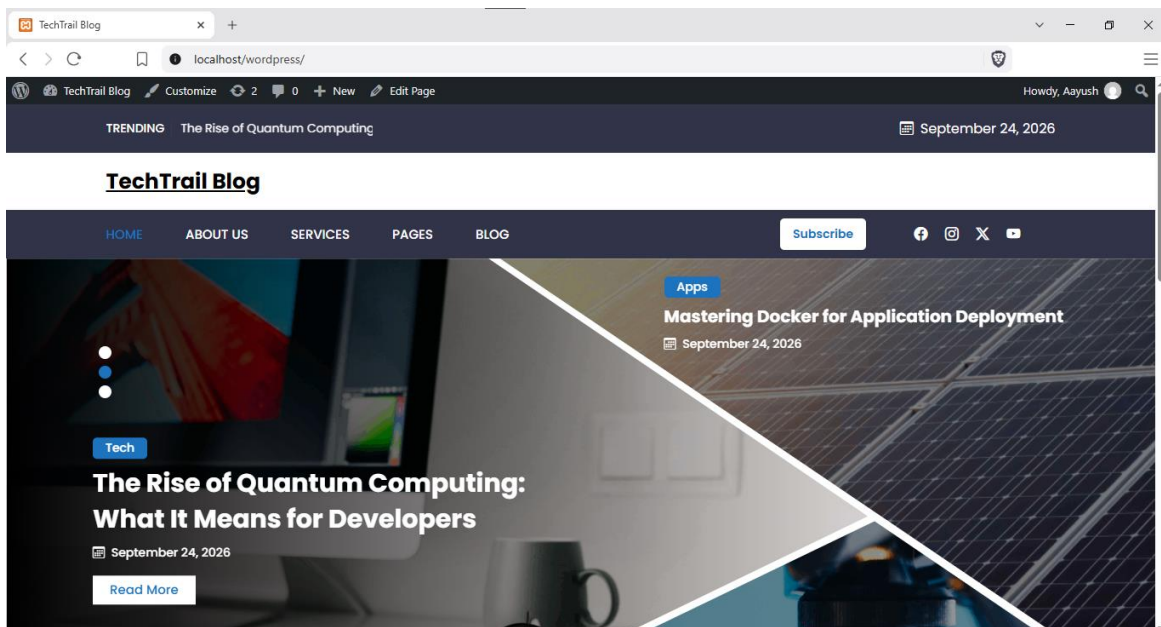


Figure 6: Installed Theme

1.5 Downloading and Installing a WordPress Plugin

Step by Step Procedure (from WordPress.org)

1. Visit WordPress.org Plugins.
2. Search for the required plugin.
3. Open the plugin and click "Download".
4. The plugin will be downloaded as a .zip file.
5. Login to the WordPress Dashboard.
6. Go to "Plugins" → "Add New Plugin".
7. Click "Upload Plugin".
8. Select the downloaded .zip file.
9. Click "Install Now".
10. After installation, click "Activate Plugin".

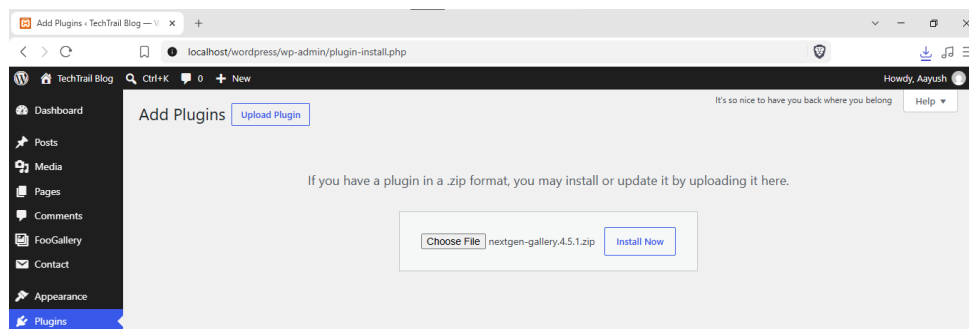


Figure 7: Adding Plugin .zip file

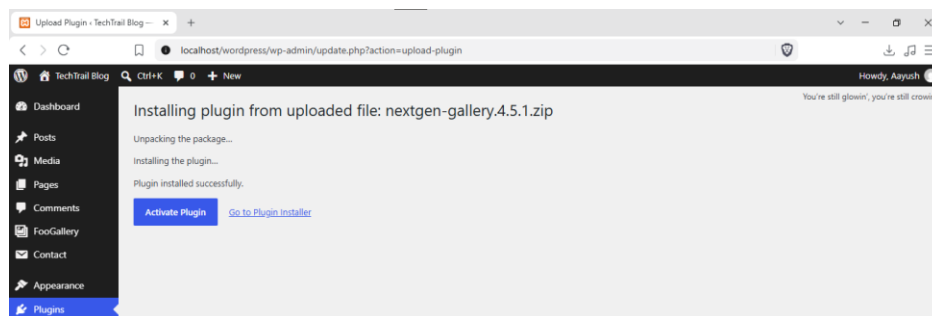


Figure 8: Installing plugin

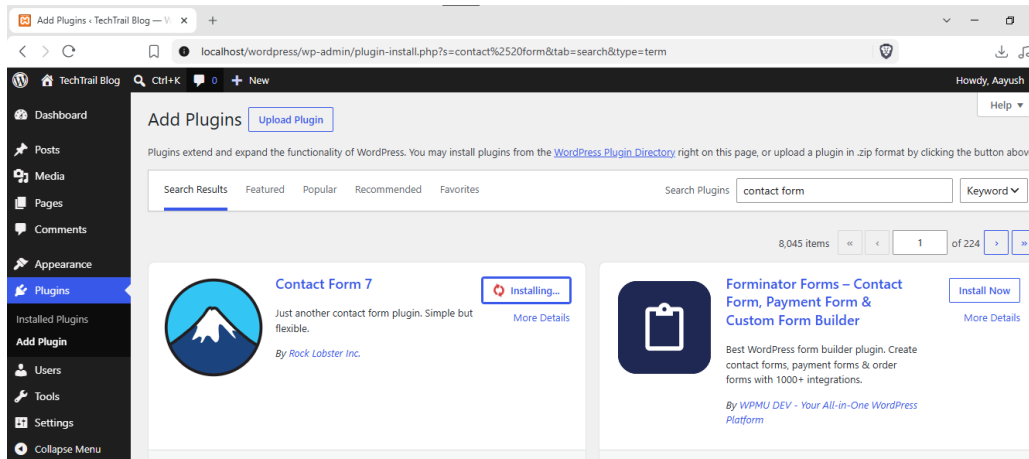


Figure 9: Installing plugin from inbuilt section

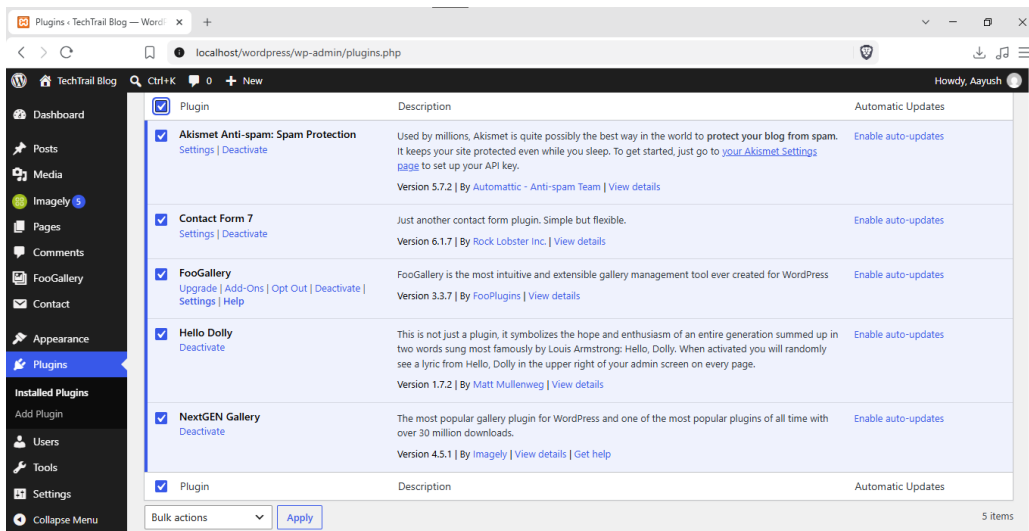


Figure 10: Activating plugins

1.6 Installing Joomla

Step by Step Procedure

1. With Apache and MySQL running in XAMPP, open phpMyAdmin (<http://localhost/phpmyadmin>) and create a new database, e.g., joomla_db.
2. Download the latest Joomla package (Full Package, .zip) from joomla.org/download.
3. Extract the ZIP file and place the contents inside a new folder in htdocs, e.g., C:\xampp\htdocs\joomlasite.
4. Open a browser and navigate to <http://localhost/joomlasite>. The Joomla Web Installer starts automatically.

Step 1. Main Configuration: enter the Site Name, Description, Admin Email, Admin Username and Admin Password, then click "Next".

Step 2. Database Configuration: select Database Type (MySQLi), Host Name (localhost), Username (root), Password (blank), Database Name (joomla_db), and Table Prefix, then click "Next".

Step 3 Overview: review the summary, optionally install sample data ("Default English (GB) Sample Data" is recommended for beginners), and click "Install".

5. Wait for the installation progress bar to complete, then click "Remove Installation Folder", Joomla will not run until the /installation directory is deleted for security reasons.
6. Click "Site" to view the front end, or "Administrator" to log in to the Joomla backend control panel using the admin credentials created earlier.

Note: Joomla will refuse to load the site at all until the /installation folder is removed, this is a mandatory security step, unlike WordPress which does not require manual folder deletion.

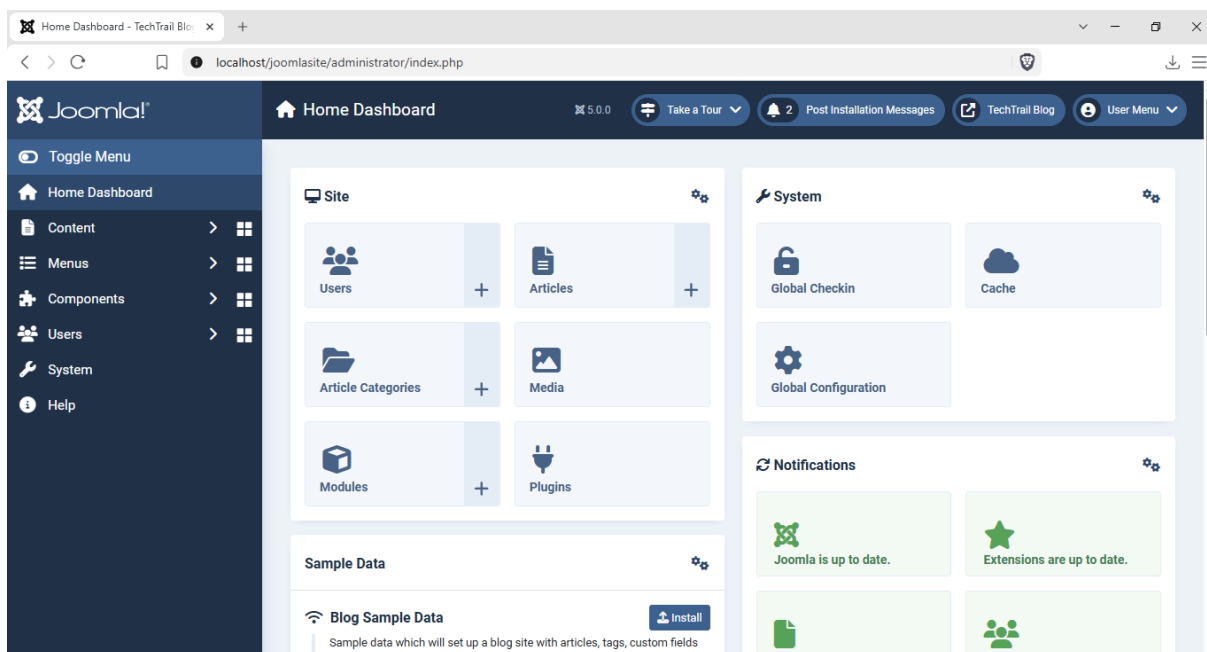


Figure 11: Joomla Dashboard

Q 16 Creating Posts and Pages

Both WordPress and Joomla distinguish between time based, list style content ("Posts" in WordPress, "Articles" in Joomla) and static content ("Pages" in both systems). Posts/Articles are typically used for blog entries and news, while Pages hold permanent content such as "About Us" or "Contact".

1.1 Creating an Article and a Page (Menu Item) in Joomla

Creating an Article

1. Log in to the Joomla Administrator panel (<http://localhost/joomlasite/administrator>).
2. From the top menu, go to "Content" → "Articles" → "New".
3. Enter the article "Title" (e.g., "Welcome to Our Site").
4. Type the article body in the TinyMCE editor provided.
5. On the right hand tabs, set the "Status" to Published, choose a "Category" (create one under Content → Categories if needed), and optionally set "Featured" to Yes so it appears on the homepage.
6. Click "Save & Close" to publish the article.

Creating a Page (via a Menu Item)

Unlike WordPress, Joomla does not have a separate "Page" content type, static pages are created by linking a Menu Item to a Single Article or a custom layout.

1. Go to "Content" → "Site Modules" is not needed here — instead navigate to "Menus" → select a menu (e.g., "Main Menu") → "New".
2. Enter the Menu Title (e.g., "About Us").
3. Under "Menu Item Type", click "Select" and choose "Articles" → "Single Article".
4. Select the article created earlier (or a dedicated one written for this static page) to link it.
5. Click "Save & Close". The item now behaves as a static page accessible from the site's navigation.

1.2 Creating a Post and a Page in WordPress

Creating a Post

1. Log in to the WordPress Dashboard (<http://localhost/mysite/wp-admin>).
2. From the left sidebar, hover over "Posts" and click "Add New".
3. Type the post title in the top field
4. Click inside the content area (Block Editor / Gutenberg) and write the post body. Use the "+" icon to add blocks such as paragraphs, images, headings, or lists.
5. On the right hand panel, set the "Category" (or create a new one) and add relevant "Tags".
6. Optionally set a "Featured Image" from the right panel.
7. Click "Preview" to check how the post will appear on the live site.
8. Click the blue "Publish" button (top right), then confirm by clicking "Publish" again.

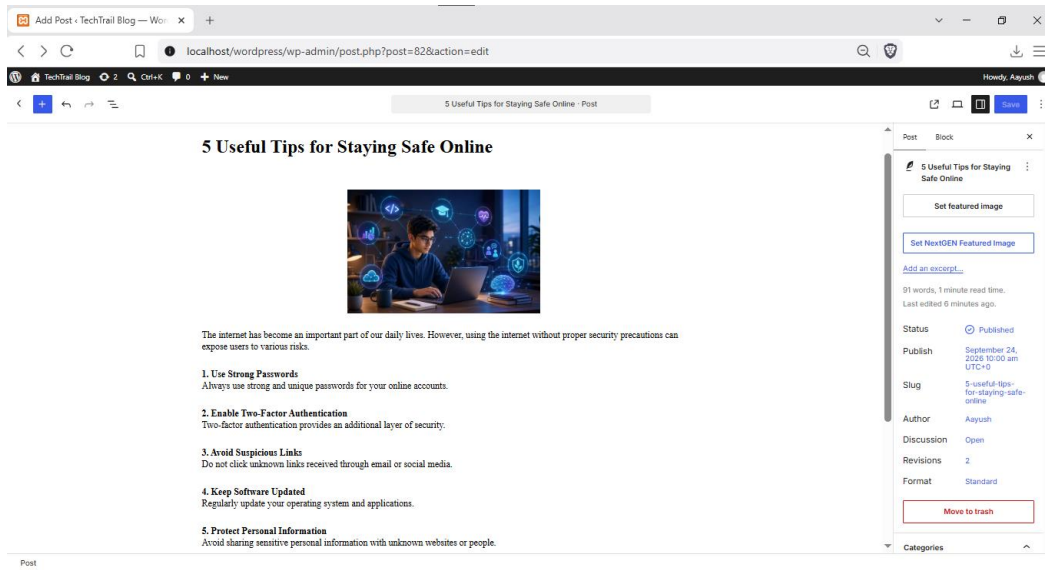


Figure 1: WordPress block editor

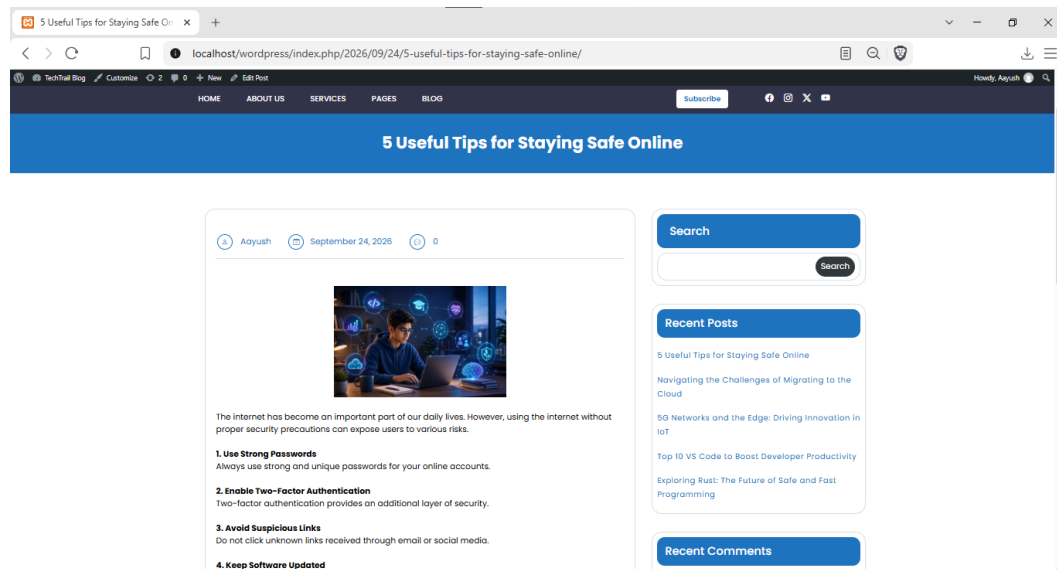


Figure 2: Published "5 Useful Tips for Staying Safe Online" post

Creating a Page

1. From the sidebar, hover over "Pages" and click "Add New".
2. Enter a title such as "About Us".
3. Add the page content using the block editor (text, images, etc.).
4. In the "Page Attributes" panel, optionally set a "Parent" page or a custom "Template" if the theme supports one.
5. Click "Publish" to make the page live.

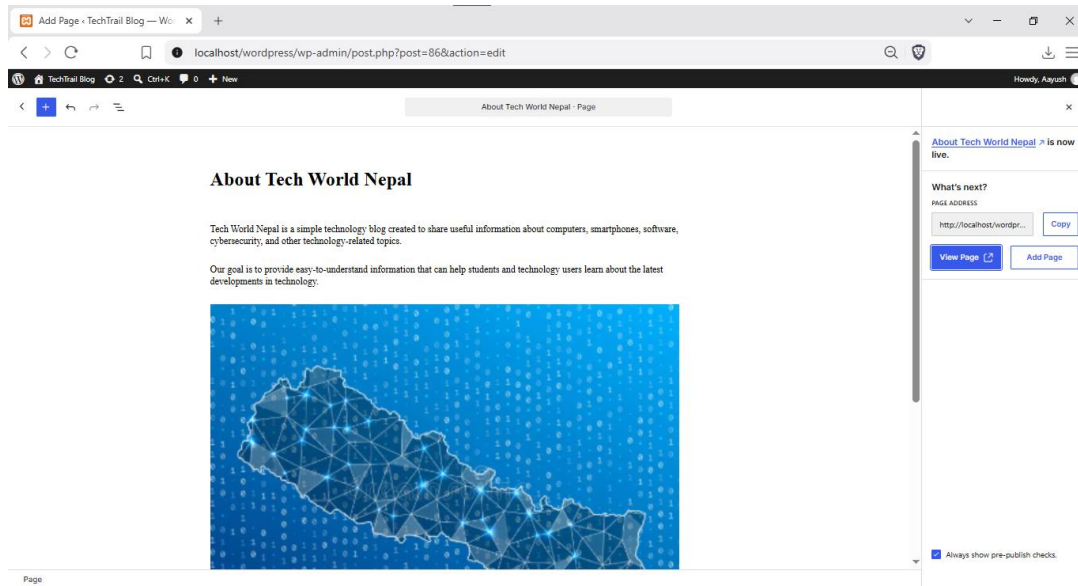


Figure 3: WordPress editor

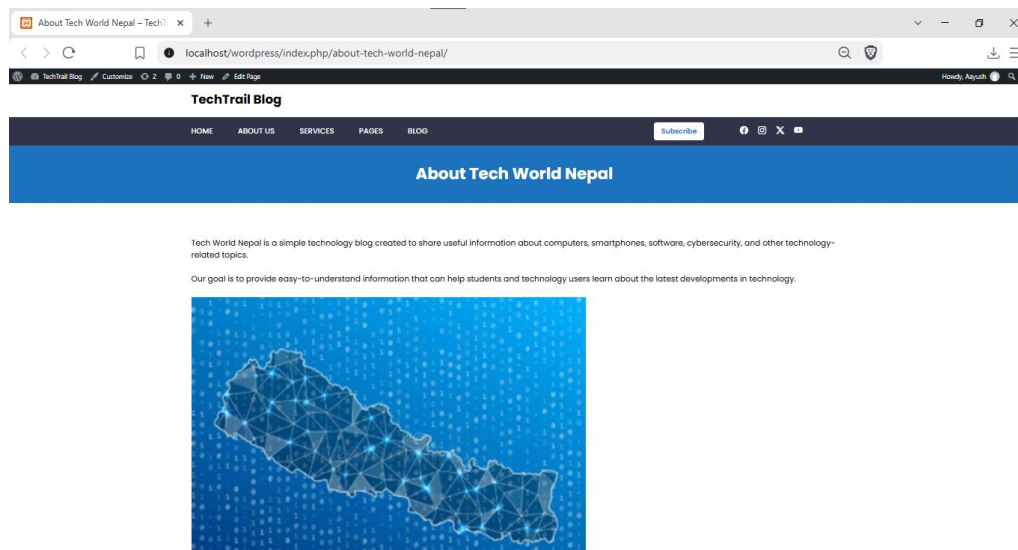


Figure 4: Published "About Tech World Nepal" page

Q 17 Creating and Managing Menus

Navigation menus allow visitors to move between different sections of a website. Both CMS platforms provide a drag and drop style menu builder, though their underlying menu architecture differs.

1.1 Creating and Managing a Menu in Joomla

1. In the Administrator panel, go to "Menus" → "Manage" → "New Menu" to create a menu container (e.g., "Main Menu"), or use the existing default "Main Menu".
2. Go to "Menus" → select the menu → click "New" to add a menu item.
3. Enter the "Menu Title" (e.g., "Home", "Blog", and "Contact").
4. Click "Select" next to "Menu Item Type" and choose the appropriate type (e.g., Articles → Category Blog for a blog listing, System Links → External URL, or Contacts → Single Contact).
5. Set the "Parent Item" field to nest the item under another (creating a dropdown), or leave it as "Menu Item Root" for a top level item.
6. Set "Status" to Published and click "Save & Close".
7. Go to "Content" → "Site Modules" and ensure a "Menu" type module exists, is Published, is assigned to the correct menu, and is set to display in a visible position (e.g., "menu" or "header") defined by the active template — this module is what actually renders the menu on the front end.

Note: In Joomla, creating menu items alone is not enough — a Menu module must be published and assigned to a template position for the menu to actually appear on the website's front end. WordPress menus, by contrast, appear automatically once assigned to a theme location.

1.2 Creating and Managing a Menu in WordPress

1. In the Dashboard, go to "Appearance" → "Menus".
2. Click "create a new menu", give it a name (e.g., "Main Navigation"), and click "Create Menu".
3. From the left hand boxes (Pages, Posts, Custom Links, Categories), select the items to include (e.g., Home, About Us, Blog, Contact) and click "Add to Menu".
4. In the menu structure area on the right, drag items to reorder them, or drag an item slightly to the right, underneath another, to create a drop down sub menu.
5. Under "Menu Settings", check the appropriate "Display location" (e.g., "Primary Menu") so the theme knows where to render it.
6. Click "Save Menu".
7. Alternatively, use "Appearance" → "Customize" → "Menus" for a live preview menu editor.

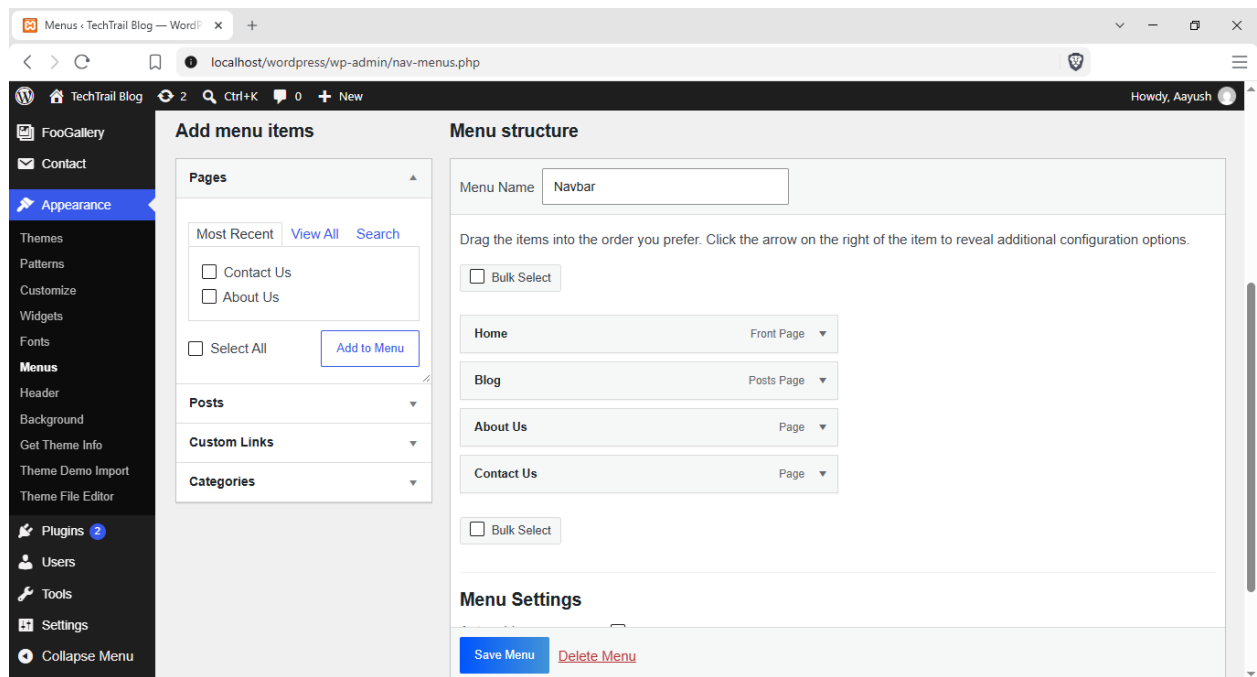


Figure 1: Menu structure

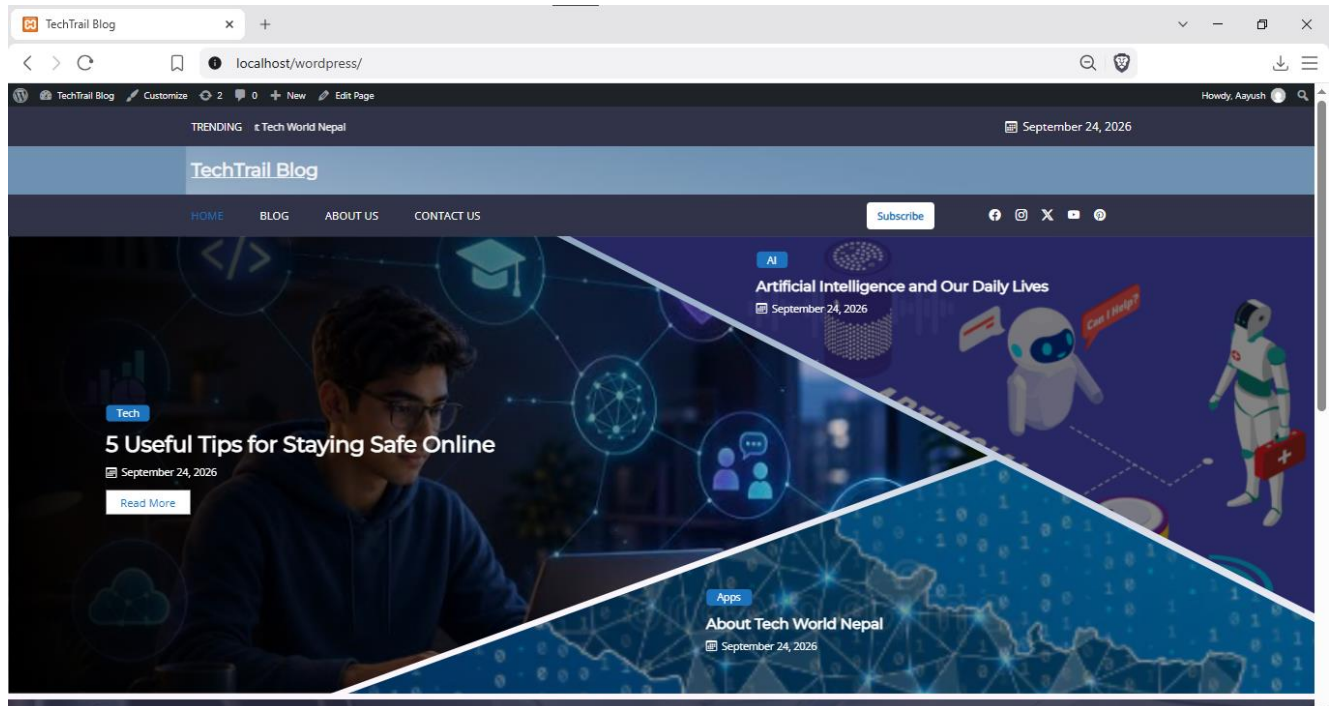


Figure 2: Menu Structure Preview

Q 18 Creating a Blog Website Using WordPress

This section documents the end to end process of building a working blog website using WordPress, combining the installation, content creation, and menu management steps covered earlier into one complete project.

1.1 Planning the Blog

- Blog Name: "TechTrail Blog"
- Purpose: Publish articles/tutorials on technology topics for a general audience.
- Pages needed: Home, Blog, About, Contact.

1.2 Step by Step Build Process

1. Install WordPress locally, following the procedure described in above labs, then log in to the Dashboard.
2. Go to "Settings" → "General" and set the Site Title to "TechTrail Blog" and add a tagline matching the blog's tech focused identity.
3. Go to "Appearance" → "Themes" → "Add New Theme" → "Upload Theme", install the "technical blogging" theme following the procedure in Section 1.4, then click "Activate".
4. Go to "Plugins" → "Add New Plugin" and install the following Contact Form 7 and other plugin, following the procedure.
5. Go to "Posts" → "Categories" and create the categories planned above: Tech, AI, and Apps.
6. Create several blog posts as described in Section 2.1, assigning each to an appropriate category (Tech, AI, or Apps), adding a featured image, and publishing them.
7. Create static pages "About Us" and "Contact Us" (using the Contact Form 7 shortcode inside the page) as shown above lab.
8. Go to "Settings" → "Reading" and set the homepage to display the latest posts through the theme's built in hero slider layout.
9. Go to "Appearance" → "Menus" and build a menu named "Navbar" containing Home, Blog, About Us, and Contact Us, then assign it to the theme's Primary Menu location as described in above labs.
10. Go to "Appearance" → "Widgets" and add useful widgets to the sidebar, such as Search, Recent Posts, and Recent Comments.
11. Preview the site on the front end (<http://localhost/mysite>) to verify that navigation, posts, pages, and widgets all display correctly.
12. Customize the header color scheme and slider images via "Appearance" → "Customize" to finish the visual design.

1.3 Final Website Structure

Menu Item	Content Type	Purpose
Home	Latest Posts / Static Front Page	Landing page introducing the blog
Blog	Post Archive	Lists all published blog posts
About	Page	Information about the site/author
Contact US	Page + Contact Form	Allows visitors to send messages

The completed website demonstrates a fully functional blog: visitors can browse categorized posts, navigate through a custom menu, read static informational pages, and get in touch through a contact form, fulfilling the core requirements of a modern blog website built on WordPress.

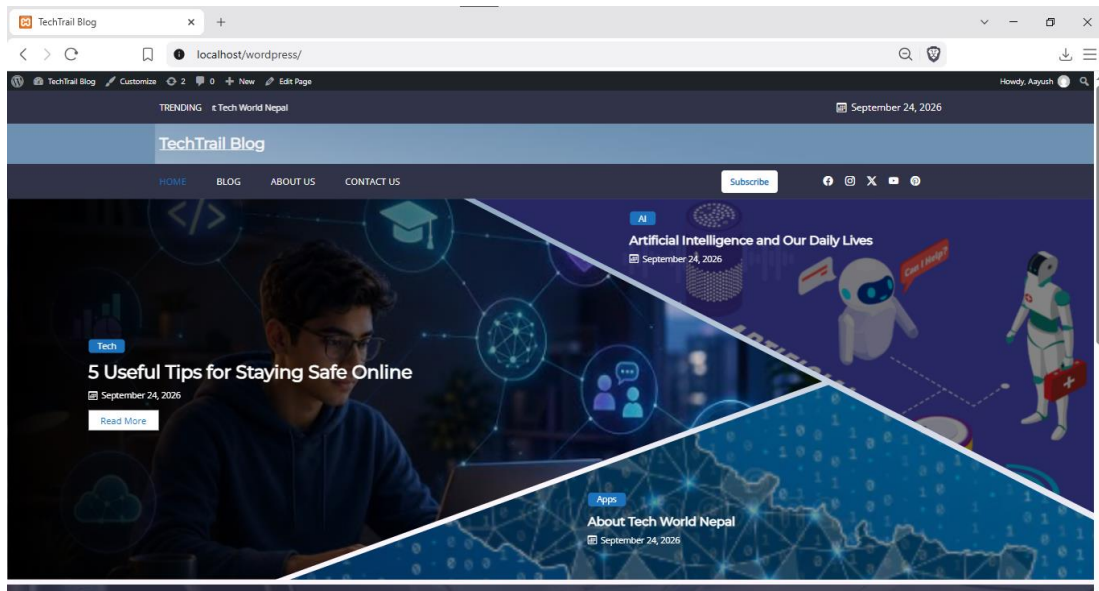


Figure 1: Final published homepage.

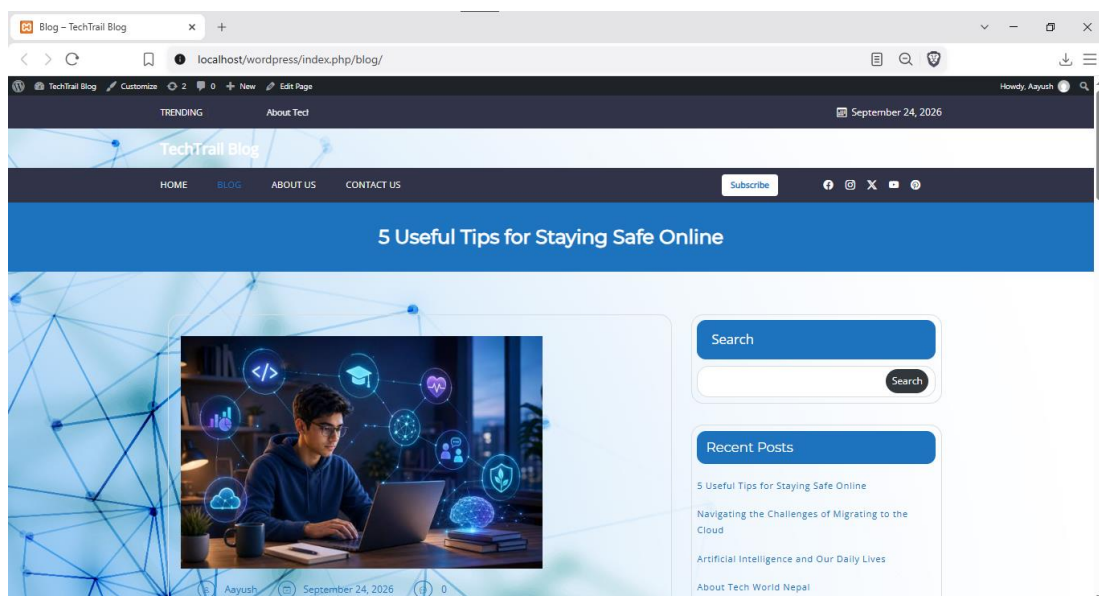


Figure 2: Final published blog

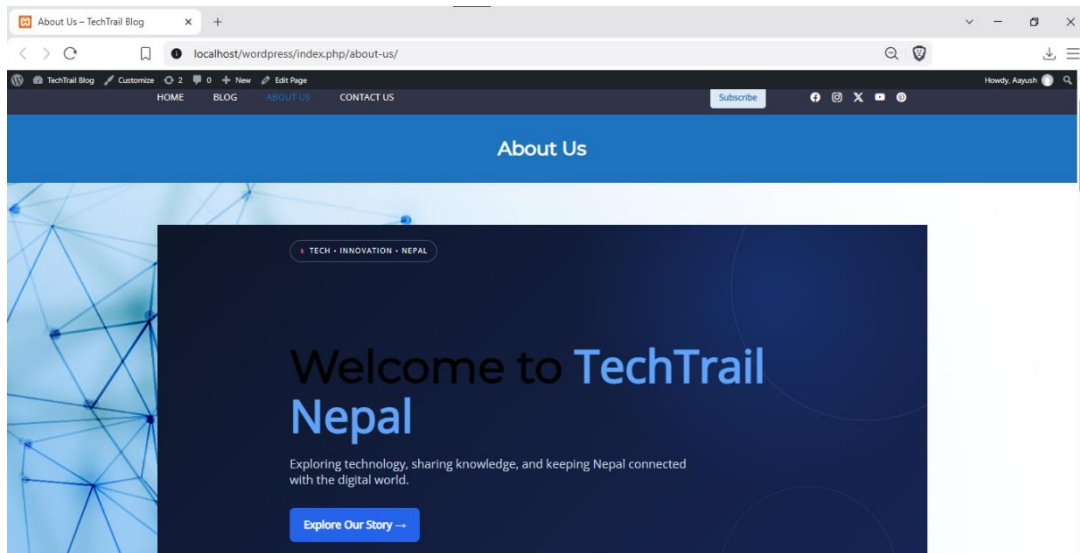


Figure 3: Final published about us

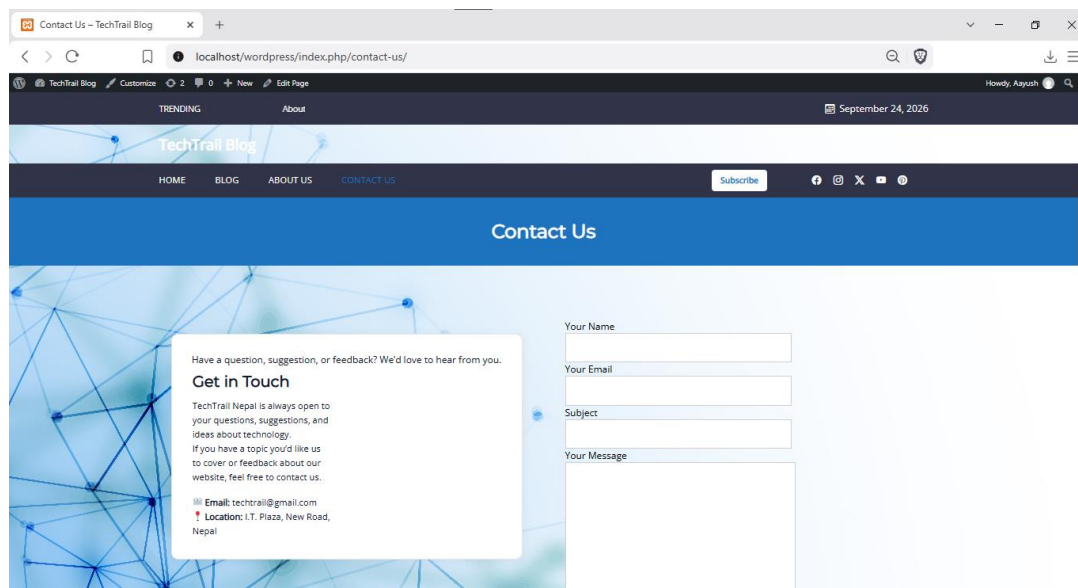


Figure 4: Final published contact us